Part 3

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Part A

```
% Givens:
f = 200e6; % [Hz] Modulation frequency
mu_a = 0.1; % [1/cm] Absorption coefficient
mu_s_p = 10; % [1/cm] Reduced scattering coefficient mu_s'
n = 1.37; % Index of refraction

% Derived constants
c_0 = 2.998e10; % [cm/s] Speed of light in vacuum
c = c_0 / n; % [cm/s] Speed of light in tissue
omega = 2*pi*f; % [rad/s] Angular frequency
D = 1 / (3*(mu_a + mu_s_p)); % [cm] Diffusion coefficient

k_r = ((c^2*mu_a^2 + omega^2) / (c^2*D^2))^(1/4) * sin(0.5*atan(omega/(c*mu_a)));
k_i = ((c^2*mu_a^2 + omega^2) / (c^2*D^2))^(1/4) * cos(0.5*atan(omega/(c*mu_a)));

lambda = 2*pi/k_r
```

```
lambda =
13.0440
```

Part B

```
% Frequency range
f_range = linspace(100e6, 1000e6, 500);% [Hz]
omega_range = 2*pi*f_range; % [rad/s]

% Vectorized versions over the frequency range (part b)
k_r_range = ((c^2*mu_a^2 + omega_range.^2) ./ (c^2*D^2)).^(1/4) .*
sin(0.5*atan(omega_range./(c*mu_a)));
k_i_range = ((c^2*mu_a^2 + omega_range.^2) ./ (c^2*D^2)).^(1/4) .*
cos(0.5*atan(omega_range./(c*mu_a)));
lambda_range = 2*pi./k_r_range
```

```
lambda_range = 1×500
    25.3959     24.9547     24.5291     24.1182     23.7212     23.3375   ···
```

```
% Plot
figure;

subplot(2,1,1);
```

```
plot(f_range/1e6, lambda_range)
ylabel("Lambda, \lambda [cm]")

subplot(2,1,2);
plot(f_range/1e6, k_i_range)
ylabel("Attentuation, k_i [1/cm]")
xlabel("Frequency, f [MHz]")
```

